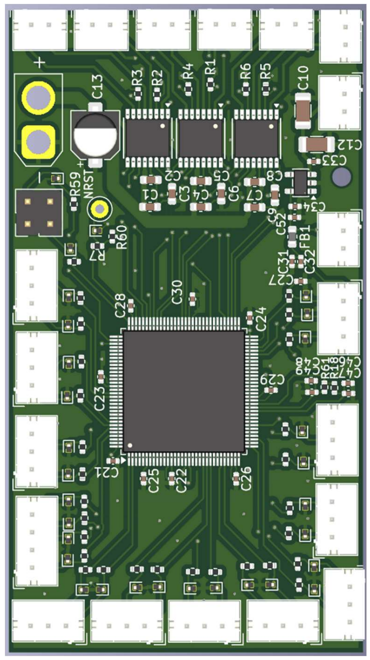
Control board v2.2



- Based on STM32G474QET6
  - 170 MHz
  - 128 Ko of SRAM
  - 512 Ko Flash
- 2 UART connections
- 9 analog lines to ADC
- 4 PWM lines
- 8 encoder inputs
- 6 motor output
- 5V and 12V (max 6A) power inputs
- 3.3V power output

Description:

A custom board made exactly for the needs of the arm prosthesis.

The board is placed in the palm, as such it has to be very compact and have few cables going to the arm

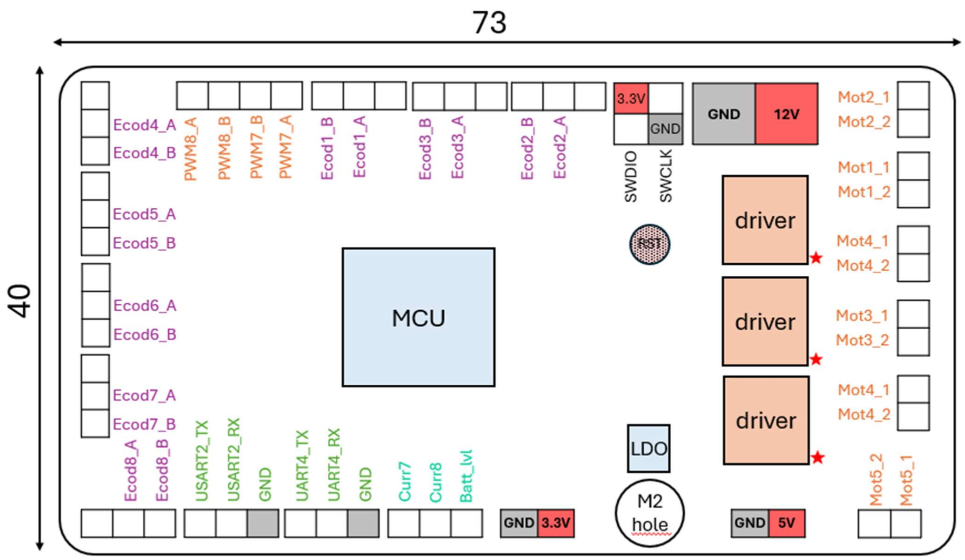
All connectors are JST for robustness and availability

Refer to this notion page for coding :

https://app.notion.com/p/n-pulse/Software-2a1e4f65ffef80c39081e30f81aa752e

Index

1) Functions and pins



a) Analog

Board PIN	MCU PIN	Function
Curr1	PA3	OPAMP1 (ADC1)
Curr2	PA7	OPAMP2 (ADC2)
Curr3	PB0	OPAMP3 (ADC2)
Curr4	PD11	OPAMP4 (ADC5)
Curr5	PE13	ADC3_IN3
Curr6	PE14	ADC4_IN1
Curr7	PC3	OPAMP5 (ADC5)
Curr8	PD9	OPAMP6 (ADC4)
Batt_lvl	PE15	ADC4_IN2

b) Encoders

Board PIN	MCU PIN	Function
Ecod1_A	PA15	TIM8_CH1
Ecod1_B	PB8	TIM8_CH2
Ecod2_A	PD3	TIM2_CH1
Ecod2_B	PD4	TIM2_CH2
Ecod3_A	PB6	TIM4_CH1
Ecod3_B	PB7	TIM4_CH2
Ecod4_A	PE2	TIM20_CH1
Ecod4_B	PE3	TIM20_CH2
Ecod5_A	PC0	TIM1_CH1
Ecod5_B	PC1	TIM1_CH2
Ecod6_A	PB5	LPTIM1_IN1
Ecod6_B	PC2	LPTIM1_IN2
Ecod7_A	PA0	TIM5_CH1
Ecod7_B	PA1	TIM5_CH2
Ecod8_A	PA6	TIM3_CH1
Ecod8_B	PA4	TIM3_CH2

c) PWM

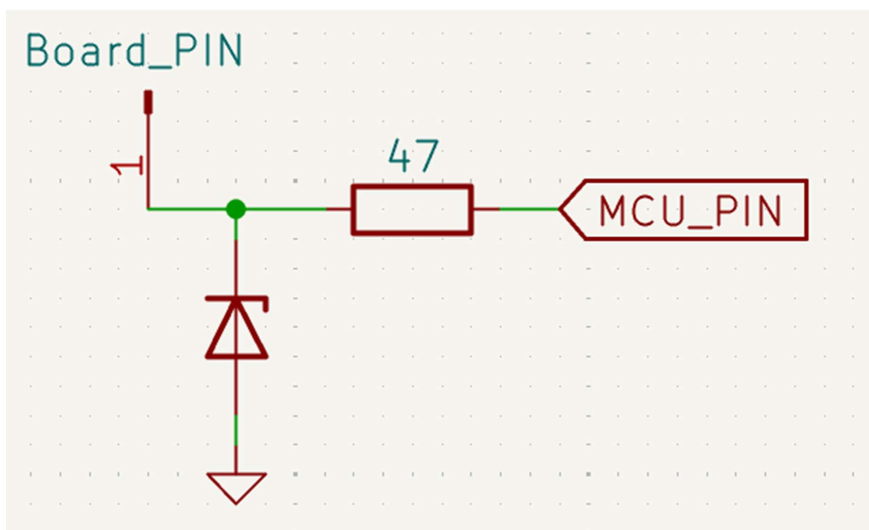
Board PIN	MCU PIN	Function
PWM1_A	PA8	HRTIM_CHA1
PWM1_B	PA9	HRTIM_CHA2
PWM2_A	PA11	HRTIM_CHB2
PWM2_B	PA10	HRTIM_CHB1
PWM3_A	PC6	HRTIM_CHF2
PWM3_B	PC7	HRTIM_CHF1
PWM4_A	PC9	HRTIM_CHE2
PWM4_B	PC8	HRTIM_CHE1
PWM5_A	PB12	HRTIM_CHC1
PWM5_B	PB13	HRTIM_CHC2
PWM6_A	PB15	HRTIM_CHD2
PWM6_B	PB14	HRTIM_CHD1
PWM7_A	PF9	TIM15_CH1
PWM7_B	PF10	TIM15_CH2
PWM8_A	PE0	TIM16_CH1
PWM8_B	PE1	TIM17_CH1

d) Communication

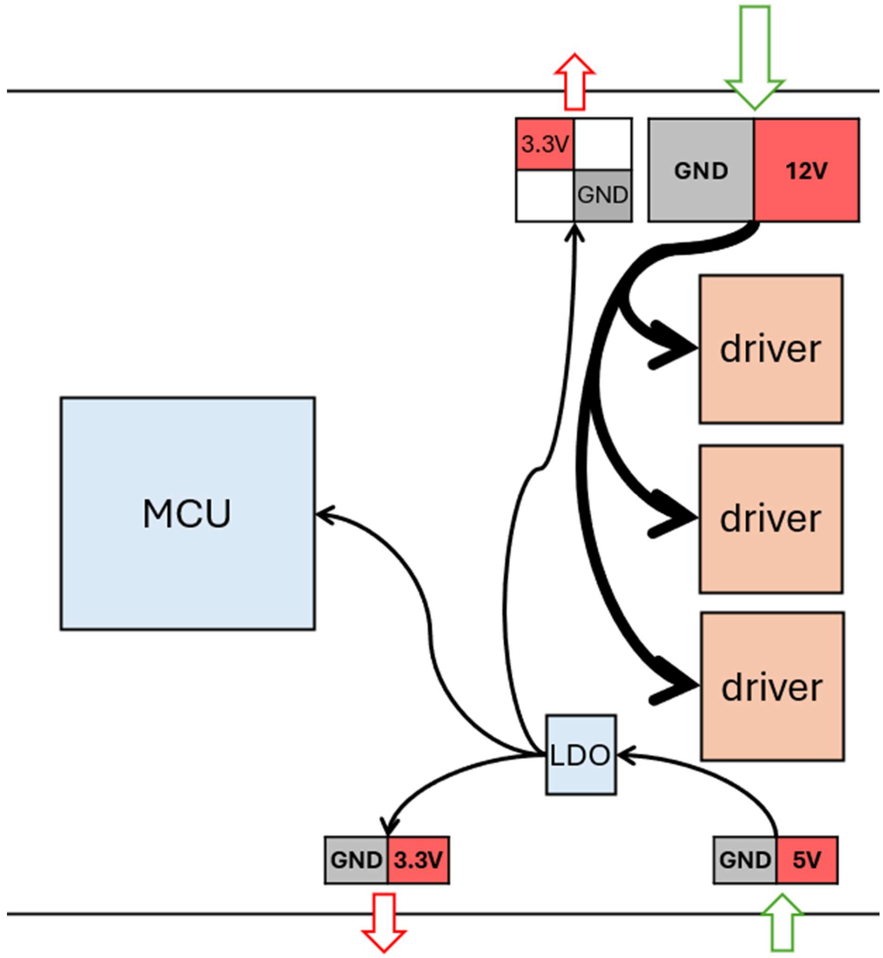
Pin on board	Pin on MCU	Function
UART_EMG_RX	PC11	UART4_RX
UART_EMG_TX	PC10	UART4_TX
UART_Hapt_RX	PD6	USART2_RX
UART_Hapt_TX	PD5	USART2_TX
SWCLK	PA14	SYS_SWCLK
SWDIO	PA13	SYS_SWDIO

e) Wiring

3.3V ESD protection with 3.6V breakdown voltage TVS diode and 47 Ohm resistor



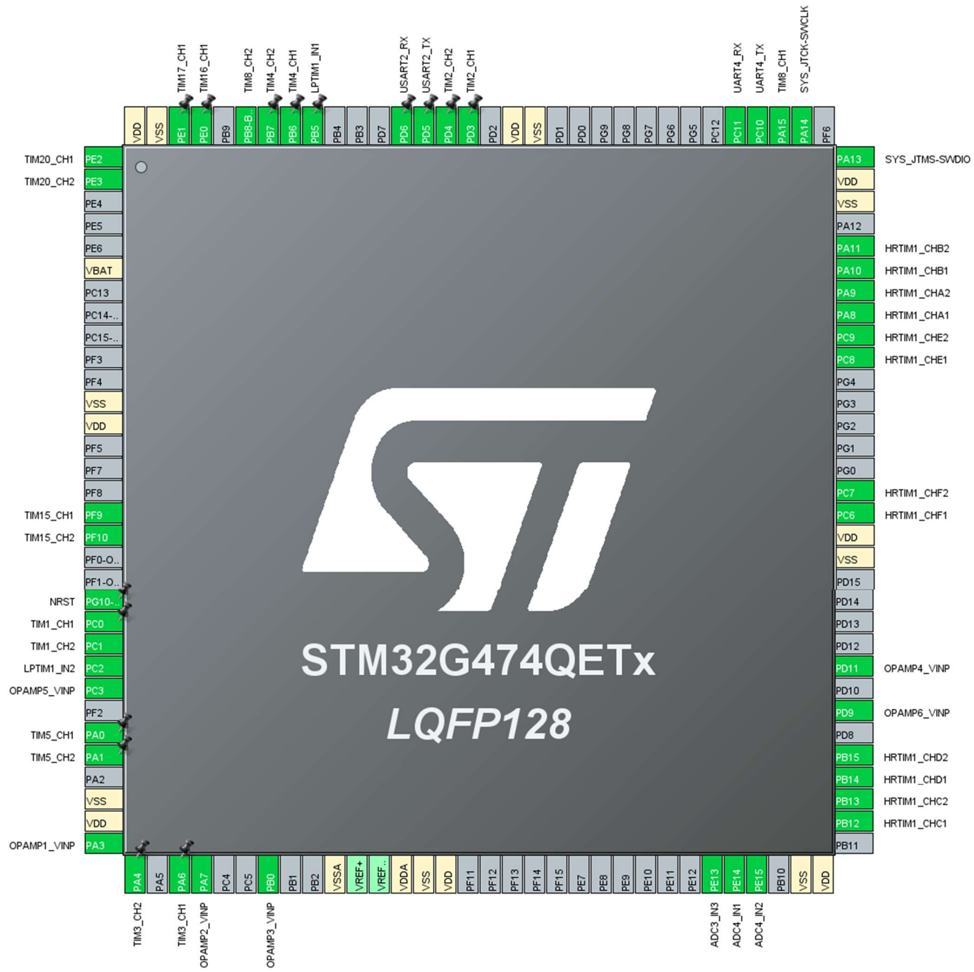
2) Power pins



12V alimentation for the motors directly to the drivers (with bulk and decoupling capacitors) max current : 6A (driver limits at 1A per motor)

5v alimentation from the power board goes through the LDO to power in 3.3V all on board logic and external elements (encoders, debugger)

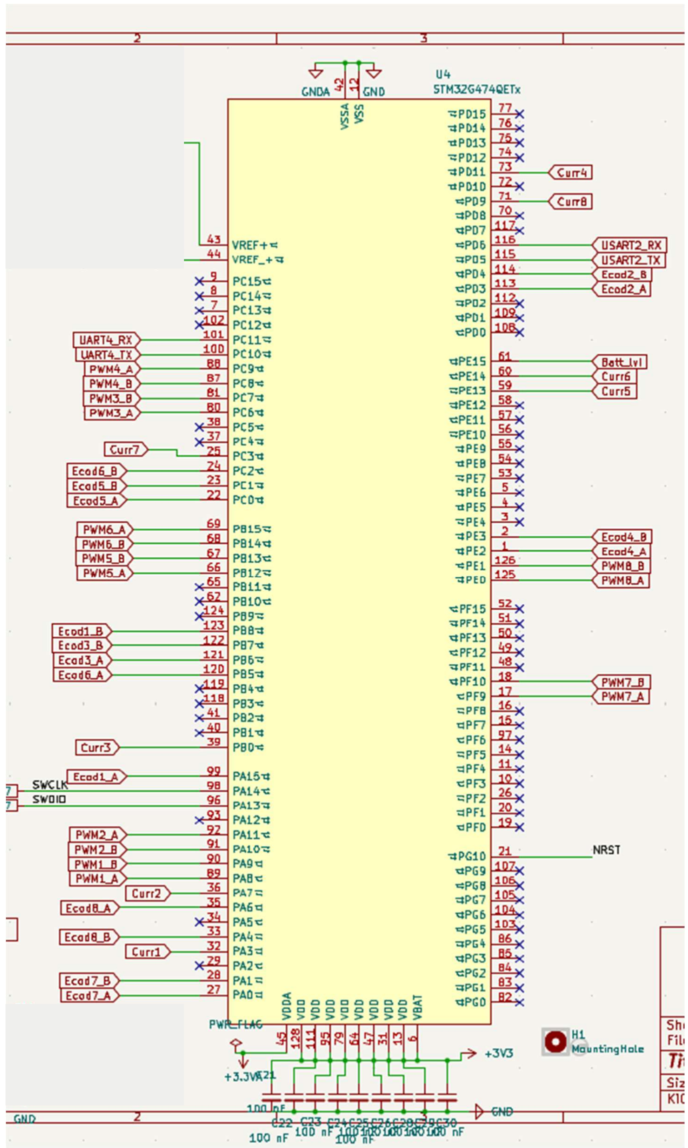
3) MCU pinout



Refer to digital and analog pinouts for details

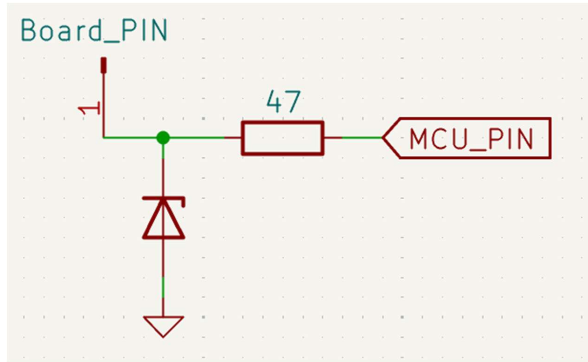
4) Schematics

a) General



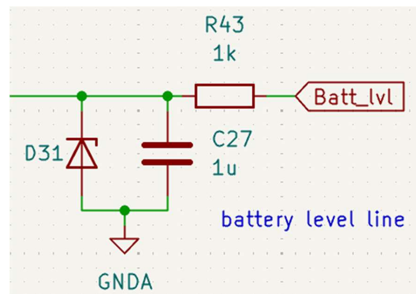
b) Analog and digital lines

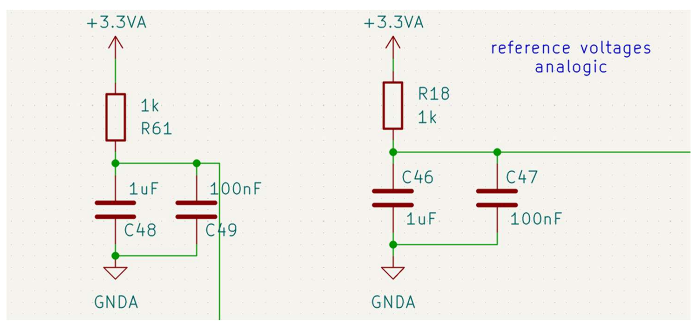
Digital :



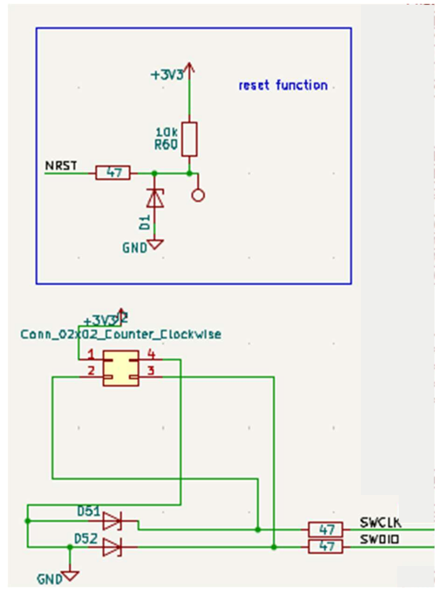
Analog:

Filtre RC a ~160Hz (only on battery line, others are like the digital lines)





c) Connectivity

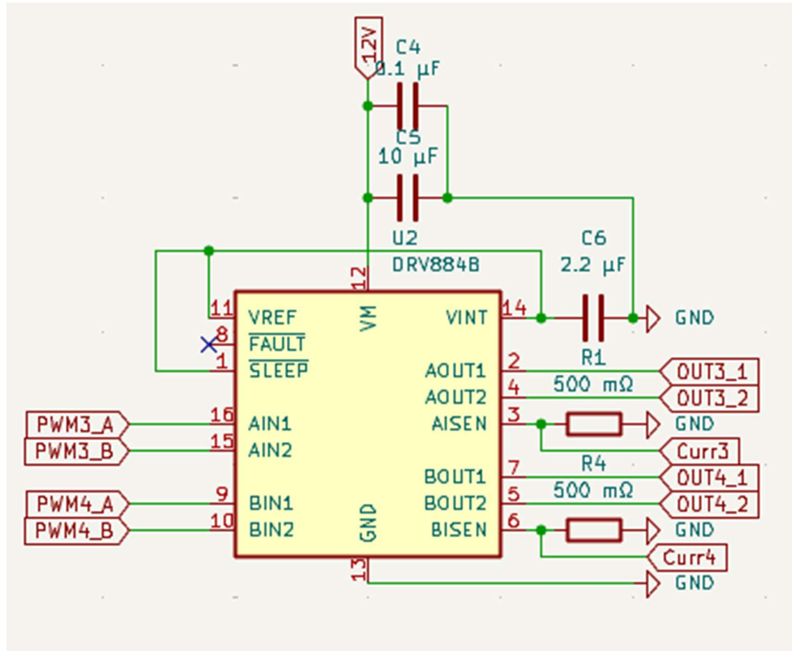


The debugger is external, we use a ST-link (for us it is the v3 minie, but any one compatible with Serial Wire debugging can be used)

We connect only the SWDIO and SWCLK in addition to power.

The whole board can be reset manually in case of wrong programming locking the Serial Wire by touching the test point to ground (warning to cover the test point during integration to ensure there is no faulty connection)

d) Drivers



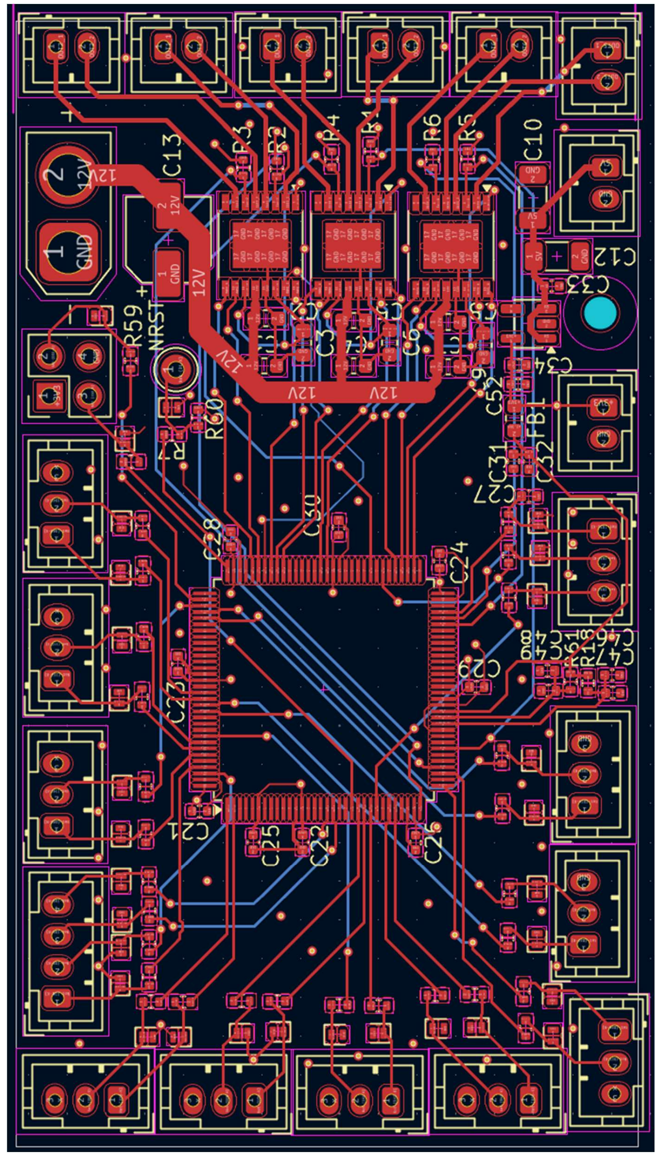
The 500 mOhm resistors tell the driver to limit the current at 1A (it does not cut, simply does not supply more). This system is not exact and we have driver consumption. The power system must be able to withstand 7A for a short time or have its own safety.

	PWM_A on	PWM_A off
PWM_B on	Braking	Turning -
PWM_B off	Turning +	Free

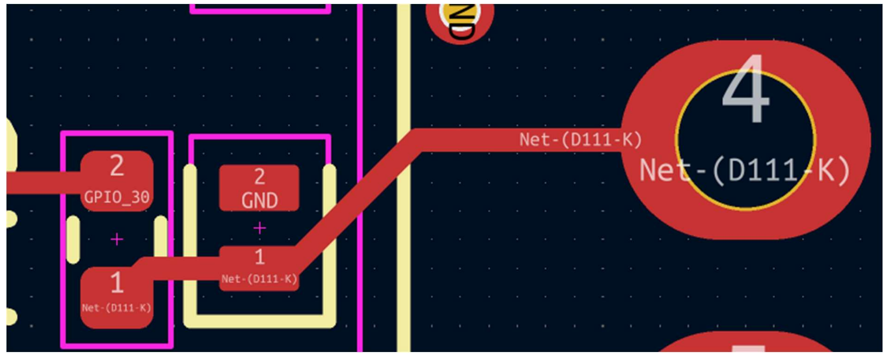
The nsleep must always be pulled high for the driver to function. For this reason, this specific board needs the user to manually pull it to high (3.3v or 5V) once before use.

5) PCB Files

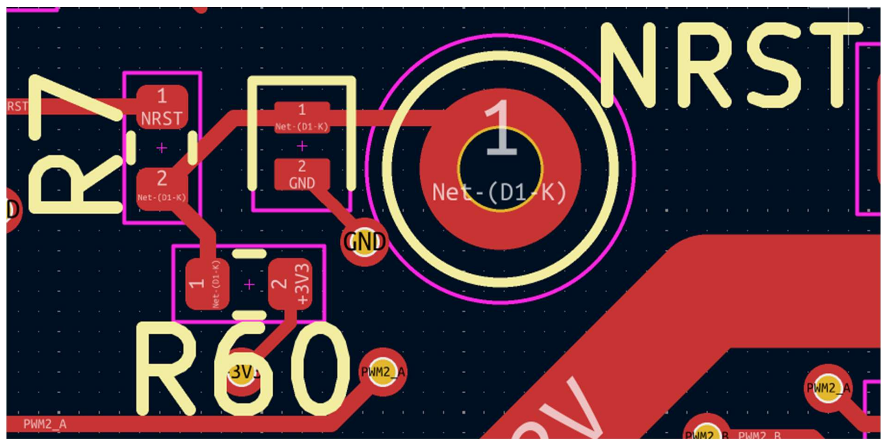
a) <u>Kicad Files</u>



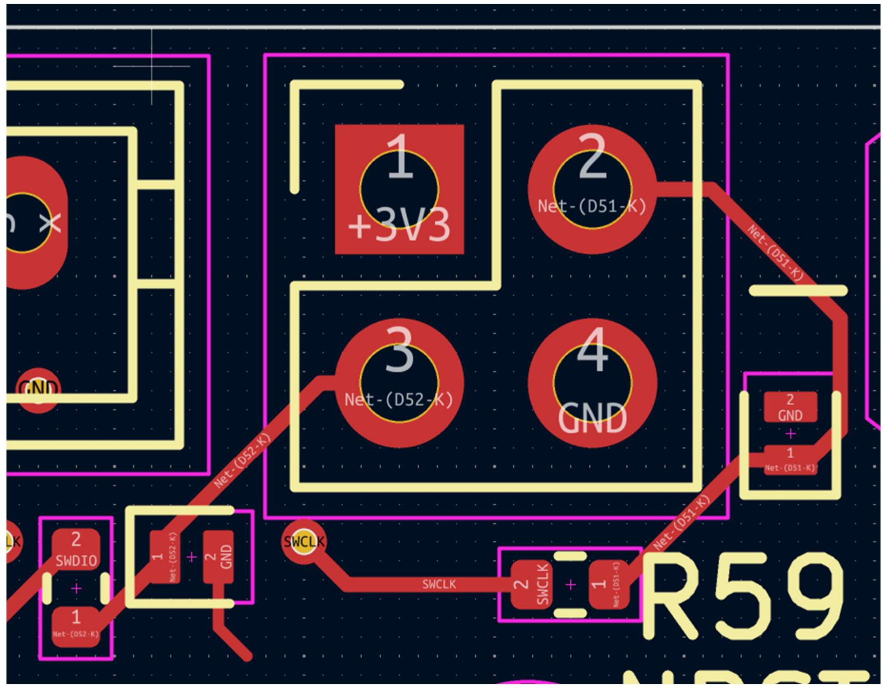
Digital trace



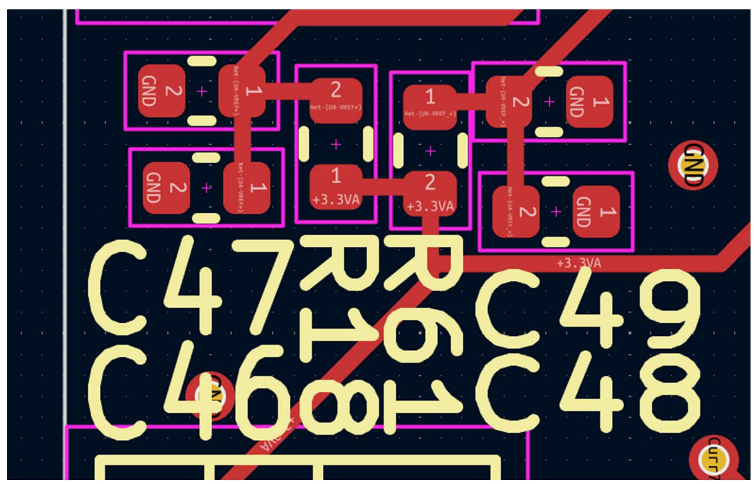
Test points



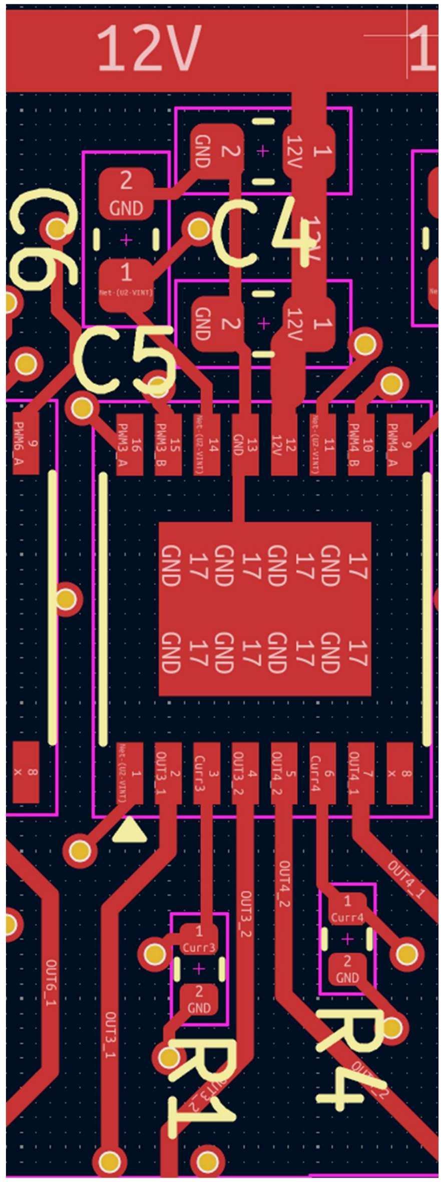
Debugger



Voltage References

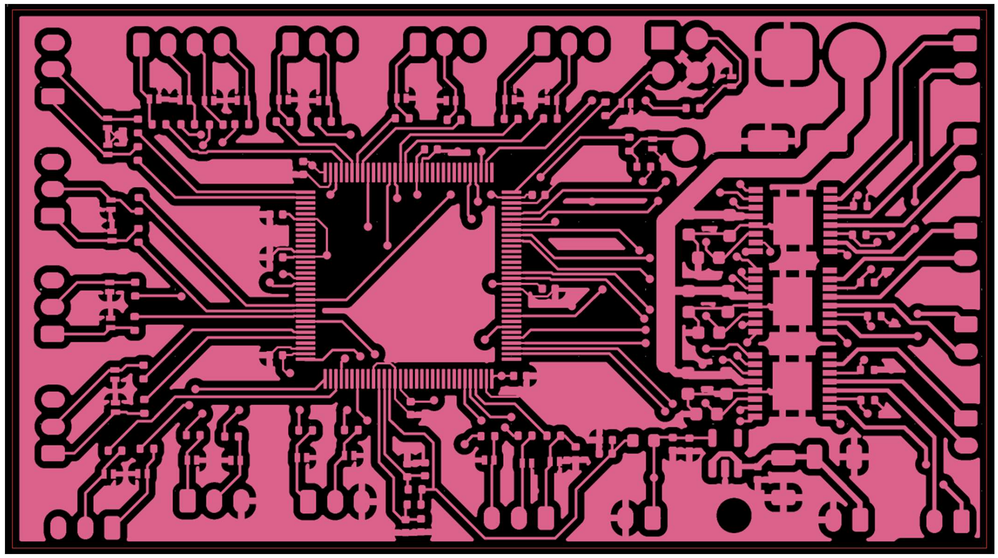


Driver module

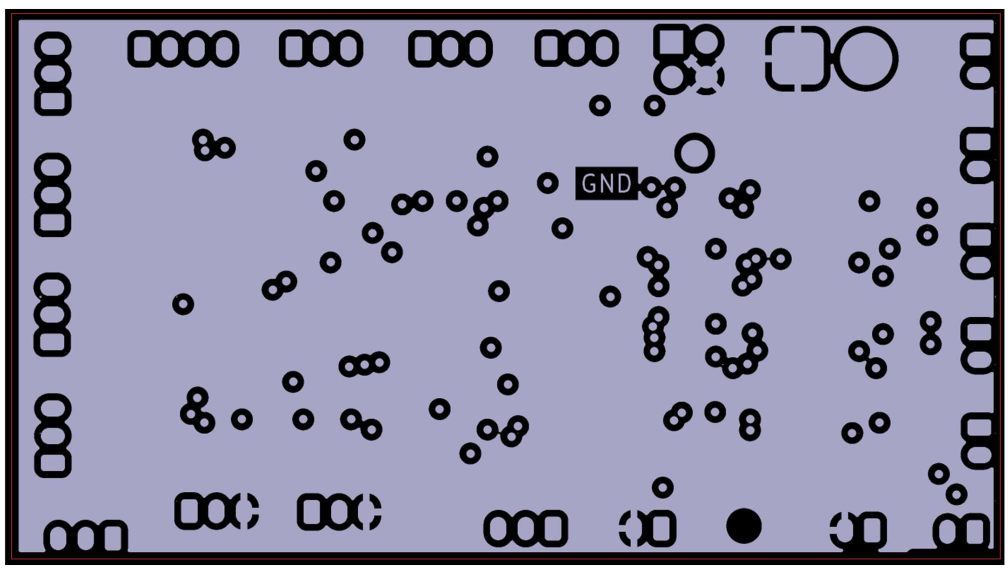


b) Gerber Files

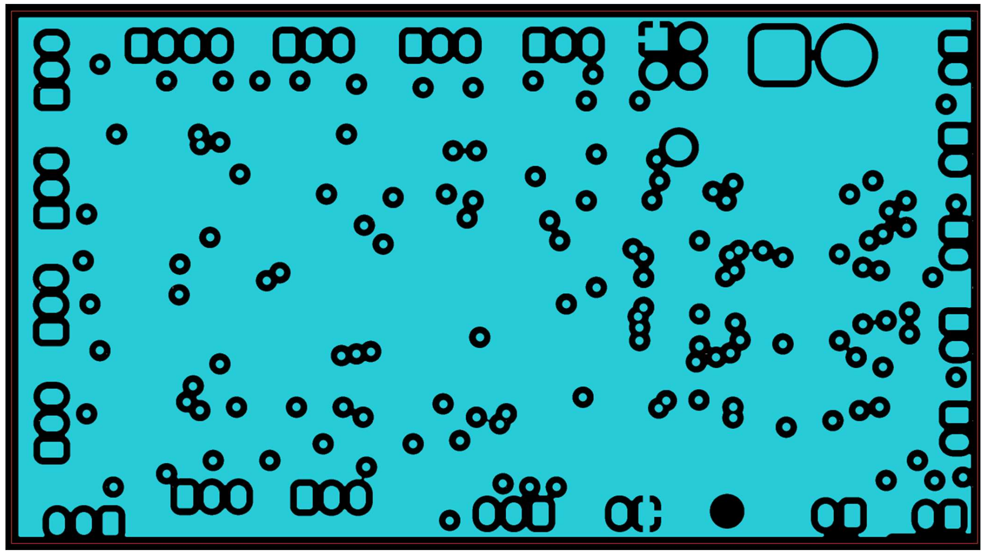
Layer 1 (data traces, 12V power, GND zone)



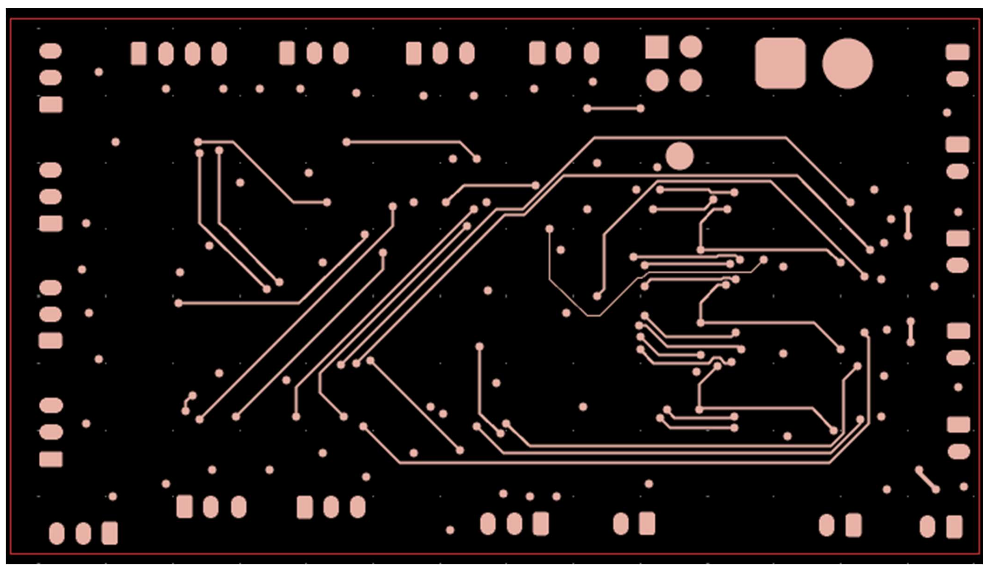
Layer 2 (GND)



Layer 3 (+3.3V)



Layer 4 (data traces, GND zone that is not visible here)



6) List of components

Connectors :

Pins : JST_PH

- 2 pins (x8)
- 3 pins (x8)
- 4 pins (x1)

Debugging connector :

- 2.54mm 2x2 header

Power connector :

- XT-30 male connector (not soldered directly on board, but at the end of cables because it is too high)

Passive components :

ESD diodes :

- DF2S5M5CT,L3F x28

Resistors (0402 in):

- 47 ohm (x28)
- 500 mOhm (x6) (250 mW)
- 1 kOhm (x3)
- 10 kOhm (x2)

Ferrite bead (0603 in)

Capacitors :

- 10 nF (0402 in)
- 100 nF (0402 in) (x15)
- 1 uF (0402 in) (x6)
- 2.2 uF (0603 in) (x3)
- 10 uF (0603 in) (x3)
- 22 uF (1206 in) (x2)
- 68 uF (electrolytic 5x5.8)

Active components :

- MCU : SMT32G474QET6
- Voltage regulator :AP2112K-3.3
- Drivers x3: DRV8848